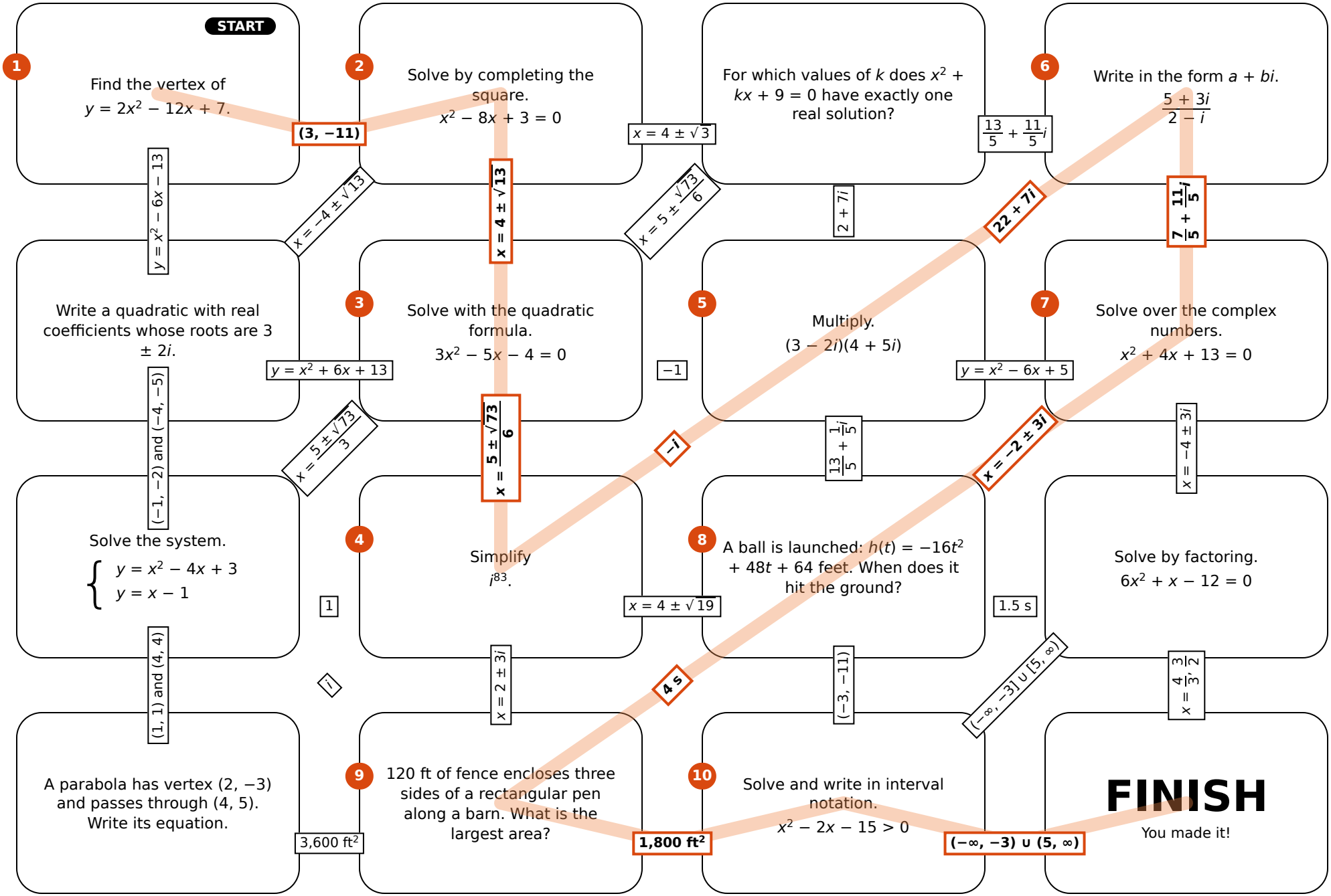
Quadratic Functions & Complex Numbers Maze **ANSWER KEY**

Name _____ Period _____

Start in the top-left room. Solve each problem. Go through the door that shows your answer. Show your work on the answer grid.



Quadratics Maze: Answer Grid **ANSWER KEY**

Name _____ Period _____

Write each answer in the order you found it, and show your work for that room.

1	Answer: **$(3, -11)$**	$x = \frac{-(-12)}{2(2)} = 3$; $2(9) - 36 + 7 = -11 \rightarrow (3, -11)$
2	Answer: **$x = 4 \pm \sqrt{13}$**	$x^2 - 8x + 16 = -3 + 16$; $(x-4)^2 = 13$; $x = 4 \pm \sqrt{13} \rightarrow x = 4 \pm \sqrt{13}$
3	Answer: **$x = \frac{5 \pm \sqrt{73}}{6}$**	$x = \frac{5 \pm \sqrt{25+48}}{6} = \frac{5 \pm \sqrt{73}}{6} \rightarrow x = \frac{5 \pm \sqrt{73}}{6}$
4	Answer: **$-i$**	$83 = 4(20) + 3$, so $i^{83} = i^3 = -i \rightarrow -i$
5	Answer: **$22 + 7i$**	$12 + 15i - 8i - 10i^2 = 12 + 10 + 7i \rightarrow 22 + 7i$
6	Answer: **$\frac{7}{5} + \frac{11}{5}i$**	multiply top and bottom by $2 + i$: $\frac{7 + 11i}{5} \rightarrow \frac{7}{5} + \frac{11}{5}i$
7	Answer: **$x = -2 \pm 3i$**	$x = \frac{-4 \pm \sqrt{-36}}{2} = \frac{-4 \pm 6i}{2} = -2 \pm 3i \rightarrow x = -2 \pm 3i$
8	Answer: **4 s**	$-16(t-4)(t+1) = 0$; $t = 4$ (time cannot be -1) $\rightarrow$ 4 s
9	Answer: **1,800 ft²**	$A = x(120 - 2x)$; vertex at $x = 30$; $A = 30(60) = 1{,}800 \rightarrow 1{,}800$ ft²
10	Answer: **$(-\infty, -3) \cup (5, \infty)$**	$(x-5)(x+3) > 0$: positive outside the roots -3 and 5; open ends $\rightarrow (-\infty, -3) \cup (5, \infty)$